INCH-POUND

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w/AMENDMENT 1
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SUPERSEDING
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PERFORMANCE SPECIFICATION SHEET

# ELECTRON TUBE, CATHODE RAY
# TYPE 12SP7D

Inactive for new design
after 26 January 2016

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Magnetic deflection and focus.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

| Parameter: | Ef | Ec1 | Ec2 | Eb | Ehk | Rg1 | Alt |
|---|---|---|---|---|---|---|---|
| Unit: | V | V dc | V dc | kV dc | V dc | MegΩ | ft |
| Maximum: | 6.9 | 0, -180 | 770 | 15 | -125 | 1/ | 60,000 |
| Minimum: | 5.7 | --- | --- | 4 | --- | --- | --- |
| Test conditions: | 6.3 | Adjust | 250 | 9 | --- | --- | --- |

See footnotes at end of table I.

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GENERAL:

Qualification: Required.

This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

AMSC N/A
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FSC 5960

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TABLE I. Testing and inspection.

| Inspection | Method MIL-STD-1311 | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|
| Qualification inspection | | | | | | |
| Base material insulating quality | 1216 | | --- | --- | --- | --- |
| Barometric pressure, reduced | 1002 | 54 mmHg | --- | --- | --- | --- |
| Cathode illumination | 5216 | | --- | --- | --- | --- |
| Direct-interelectrode capacitance | 1331 | Ck to all<br>Cg1 to all<br>Cg2 to all | Ck<br>Cg1<br>Cg2 | ---<br>---<br>--- | 8<br>10<br>10 | pF<br>pF<br>pF |
| Pressure (implosion) | 1141 | | --- | --- | --- | --- |
| Vibration (cathode-ray Tubes) | 5111 | | Width | --- | 2 | mm |
| Conformation inspection part 1 | | | | | | |
| Bulb, screen, and faceplate quality | 5106 | | --- | --- | --- | --- |
| Heater current | 1301 | | If | 540 | 660 | mA |
| Voltage breakdown | 5201 | | --- | --- | --- | --- |
| Voltage breakdown (magnetic types) | 5201 | | --- | --- | --- | --- |
| Gas ratio | 5206 | 2/ | Gr | --- | 0.25 | --- |
| Neck straightness | --- | 3/ | --- | --- | --- | --- |
| Cathode image | --- | 4/ | --- | --- | 20 | % |
| Stray light emission (conventional types) | 5216 | Eb = 15,000 V dc;<br>Ec2 = 770 V dc | --- | --- | --- | --- |
| Spot position (magnetic deflection) | 5231 | 5/ | --- | --- | 20 | mm |
| Zero-bias anode current (magnetic deflection) | 5236 | | --- | --- | --- | --- |
| Grid cutoff voltage | 5241 | | Eco | -27 | -63 | V dc |
| Grid No. 1 leakage current | 5251 | | --- | --- | --- | --- |
| Grid No. 2 leakage current | 5251 | | --- | --- | --- | --- |
| Aperture alignment | --- | 6/ | Distance | --- | 8 | mm |

See footnotes at end of table.



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TABLE I. Testing and inspection - Continued.

| Inspection | Method MIL-STD-1311 | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|
| Conformance inspection, part 2 | | | | | | |
| Electrode current (grid No. 2 current) | 5201 | Ec1 = 90 | Ic2 | --- | 50 | µA dc |
| Side terminal and base Alignment | 5101 | Pin No. 3 | --- | --- | --- | --- |
| Base pin solder depth | 1111 | | --- | --- | --- | --- |
| Neck and bulb alignment (magnetic types) | 5101 | | --- | --- | .25 | Inch |
| Face tilt | 5101 | | --- | --- | --- | --- |
| Modulation | 5223 | Ib = 200 µA dc | ΔEc1 | --- | 35 | V dc |
| Screen | 5221 | | --- | --- | --- | --- |
| Line width A (magnetic deflection) | 5226 | Ib = 200 µA dc | Width | --- | 0.38 | mm |
| Line width C (magnetic deflection) | 5226 | Ib = 200 µA dc | Width | --- | 0.60 | mm |
| Focusing ampere turns | 5246 | Ib = 200 µA dc;<br>D = 3.25 inches | AT | 450 | 550 | --- |
| Heater-cathode leakage Current | 5251 | | --- | --- | --- | --- |
| Secureness of base, cap, or insert | 1101 | | --- | --- | --- | --- |
| Permanence of marking | 1105 | | --- | --- | --- | --- |
| Conformance inspection, part 3 | | | | | | |
| Life-test provisions | --- | Group C; t = 500 hours (min);<br>Eb = 13,200 V dc;<br>Ec2 = 770 V dc;<br>Ib = 60 µA dc | --- | --- | --- | --- |
| Life-test end points | --- | Ib = 200 µA dc;<br>Line width A<br>Line width B<br>Modulation<br>Heater-cathode leakage current<br>Grid No. 1 leakage current<br>Grid No. 2 leakage current<br>Stray light emission (conventional types) | <br>Width<br>Width<br>ΔEc<br>---<br>---<br>---<br>--- | <br>---<br>---<br>---<br>---<br>---<br>---<br>--- | <br>0.38<br>0.60<br>35<br>---<br>---<br>---<br>--- | <br>mm<br>mm<br>V dc<br>---<br>---<br>---<br>--- |

See footnotes at top of next page.



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TABLE I. Testing and inspection - Continued.

1/ When Ec2 is greater than 330 V dc, Rg1 shall not exceed 0.5 Meg. When Ec2 is less than 330 V dc, Rg1 shall not exceed 1.5 Meg.

2/ This test to be performed at the conclusion of the holding period.

3/ The neck and base straightness shall be determined by the insertion of the tube neck in a cylinder 5 inches (127.00 mm) long and 1.5 inches (38.10 mm) maximum inside diameter. This cylinder shall move freely between the reference line and the base of the assembled tube.

4/ The test conditions shall be Eb = 7 kV, Ec2 = 250, focus coil current = 0, Ec1 shall be varied until the focused image of the cathode is obtained. The percentage of dead area shall be that portion of the image which is black in respect to the general brilliant area.

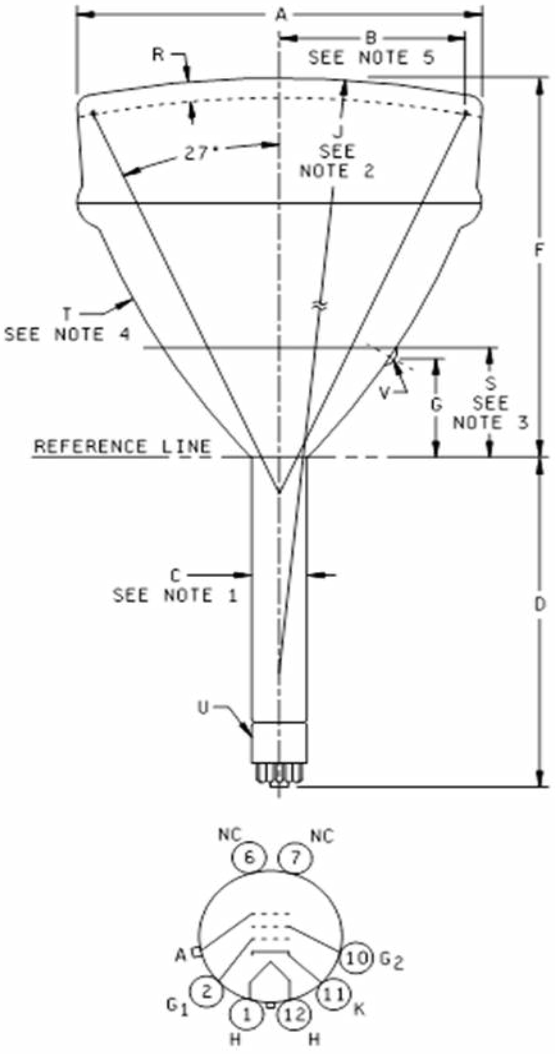
5/ There shall be no displacement due to hand capacitance or external electrostatic field.

6/ The distance between the center of the unfocused, undeflected spot at low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified herein. Ec1 shall not be held at zero for more than approximately 30 seconds to prevent damage to the screen.



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| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| | Conformance, | inspection | inspection, | part 2 |
| A | 12.340 | 12.500 | 313.44 | 317.50 |
| B | 5.500 | --- | 139.70 | --- |
| C | 1.384 | 1.500 | 35.16 | 38.10 |
| D | 8.060 | 8.440 | 204.72 | 214.38 |
| F | 10.310 | 10.690 | 261.87 | 271.53 |
| G | 2.630 | 3.380 | 66.80 | 85.85 |
| R | .300 | .400 | 7.62 | 10.16 |
| S | 2.970 | 3.090 | 75.44 | 78.49 |
| | Conformance | inspection, | part 3 | (periodic check) |
| T | Bulb: J-99 | 1/2 A1, clear | or filter | glass |
| U | Base: B7-51 | or B5-57 | | |
| V | Contact: | J1-21 | | |
| | Reference | dimensions | (see) | note 6 |
| J | 40.000 R | | 1016.00 R | |



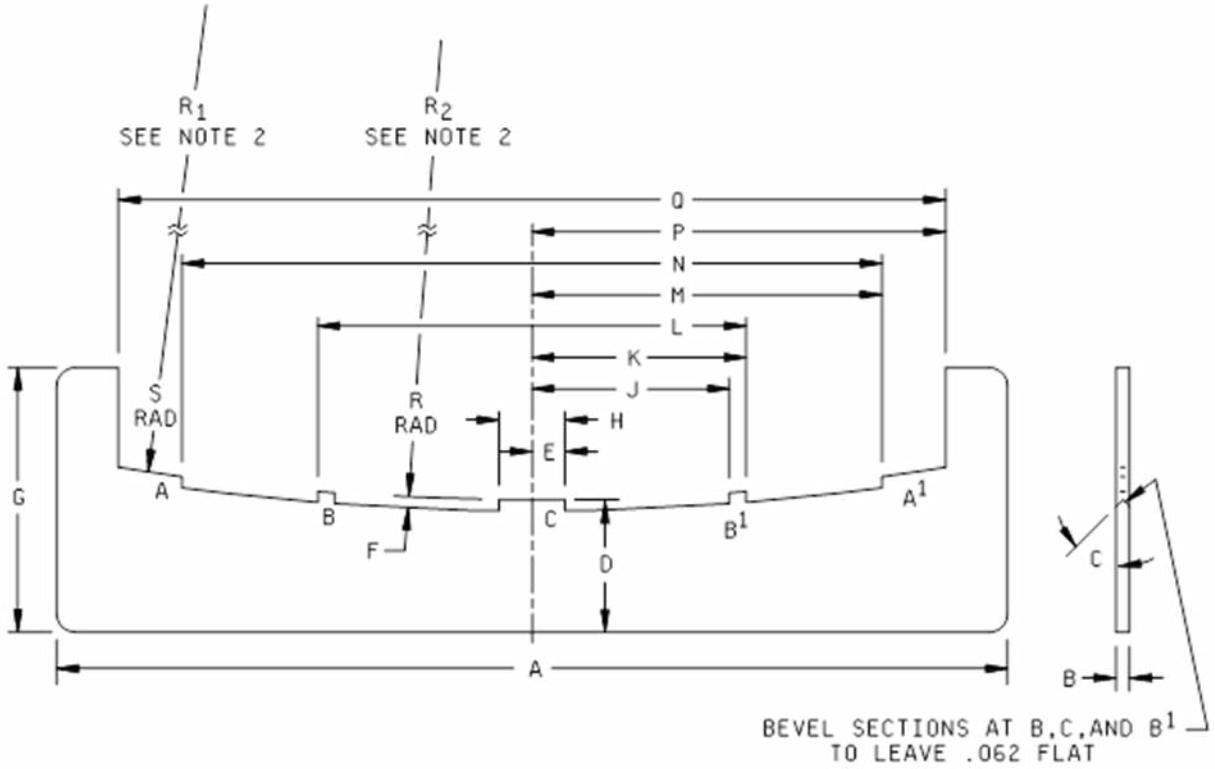

NOTES:

1. Dimension C applies to the entire length of neck below reference line. Reference line shall be determined by position where gauge 1.500 inches (38.10 mm) ± .003 inch (0.08 mm) I. D. and 2.000 inches (50.80 mm) long will rest on bulb cone.
2. To meet the requirements of faceplate gauge as shown on figure 2.
3. Line established by ring gauge of I. D. 7.188 inches (182.58 mm) ± .016 inch (0.41 mm).
4. To meet the requirements of bulb contour gauge as shown on figure 3.
5. Useful screen radius.
6. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 12SP7D.



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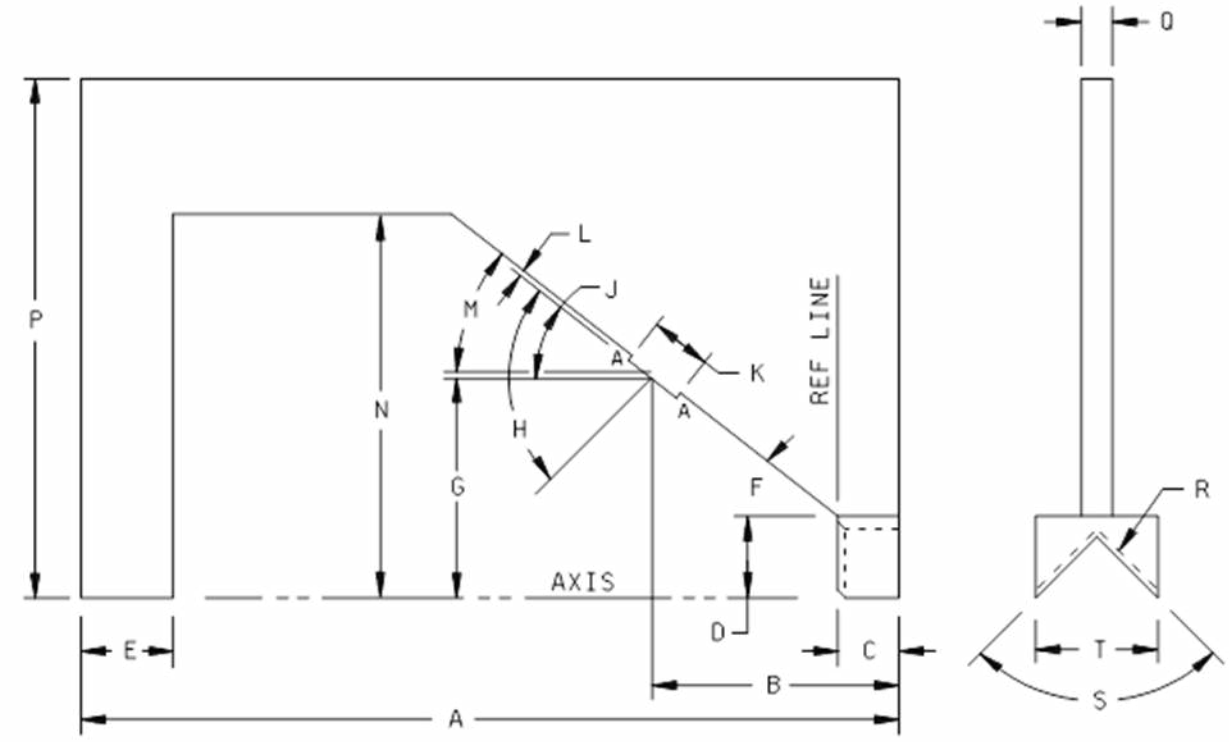

| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Ltr | Inches | | Millimeters | | Ltr | Inches | | Millimeters | |
| | Min | Max | Min | Max | | Min | Max | Min | Max |
| A | 14.187 | 14.687 | 360.35 | 373.05 | K | 3.242 | 3.258 | 82.35 | 82.75 |
| B | .115 | .135 | 2.92 | 3.43 | L | 6.484 | 6.516 | 164.69 | 165.51 |
| C | 44°30' | 45°30' | 44°30' | 45°30' | M | 5.304 | 5.320 | 134.72 | 135.13 |
| D | 1.750 | 2.250 | 44.45 | 57.15 | N | 10.609 | 10.641 | 269.47 | 270.28 |
| E | .492 | .508 | 12.50 | 12.90 | P | 6.273 | 6.289 | 159.33 | 159.74 |
| F | .052 | .072 | 1.32 | 1.83 | Q | 12.546 | 12.578 | 318.67 | 319.48 |
| G | 3.750 | 4.250 | 95.25 | 107.95 | R | 39.875 R | 40.125 R | 1012.83 R | 1019.18 R |
| H | .984 | 1.016 | 24.99 | 25.81 | S | 39.875 R | 40.125 R | 1012.83 R | 1019.18 R |
| J | 2.992 | 3.008 | 76.00 | 76.40 | | | | | |

NOTE
1. All dimensions are in inches.
2. R2 shall be equal to the absolute value of R1 ± .050 inch (1.27 mm) ± .002 inch (0.05 mm).
3. Instructions for use: Contour of face shall be checked with gauge at two orientations approximately 90 degrees. Reject if gap between bulb face and gauge points B, $B^1$, or C is greater than .100 inch (2.54 mm). Reject if bulb does not seat on steps A and $A^1$.

FIGURE 2. Faceplate gauge.



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| Ltr | Dimensions | | | | Ltr | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Inches | | Millimeters | | | Inches | | Millimeters | |
| | Min | Max | Min | Max | | Min | Max | Min | Max |
| A | 13.370 | 13.380 | 339.60 | 339.85 | L | .092 | .096 | 2.34 | 2.44 |
| B | 4.029 | 4.033 | 102.34 | 102.44 | M | 35° | 35° 30' | 35° | 35° 30' |
| C | .995 | 1.005 | 25.27 | 25.53 | N | 6.271 | 6.291 | 159.28 | 159.79 |
| D | 1.015 | 1.019 | 25.78 | 25.88 | P | 8.495 | 8.505 | 215.77 | 216.03 |
| E | 1.495 | 1.505 | 37.97 | 38.23 | Q | .495 | .505 | 12.57 | 12.83 |
| F | 37° 30' | 38° 30' | 37° 30' | 38° 30' | R | .120 x 44° 30' CHAM | .130 x 45° 30' CHAM | 3.05 x 44° 30' CHAM | 3.30 x 45° 30' CHAM |
| G | 3.592 | 3.596 | 91.24 | 91.34 | | | | | |
| H | 89° 30' | 90° 30' | 89° 30' | 90° 30' | S | 89° 30' | 90° 30' | 89° 30' | 90° 30' |
| J | 35° 30' | 36° 30' | 35° 30' | 36° 30' | T | 1.995 | 2.005 | 50.67 | 50.93 |
| K | .995 | 1.005 | 25.27 | 25.53 | | | | | |

NOTE
1. All dimensions are in inches.
2. Unless otherwise specified, tolerances shall be ± .016 inch (0.041 mm) on fractions ± .005 inch (0.13 mm) on decimals, and ± .5° on angles.
3. Instructions for use: Reject if bulb does not rest on flat 'A-A' when bulb neck is seated in V block.

FIGURE 3. Bulb contour gauge.



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Referenced documents. In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311.

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Review activities:
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